



Service Bulletin Number: 4021600

Released Date: 10-Sep-2014

Aftertreatment Diesel Oxidation Catalyst (DOC) and Aftertreatment Diesel Particulate Filter (DPF) Reuse Guidelines

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I. Introduction

- This document provides reuse guidelines for the aftertreatment diesel oxidation catalyst (DOC) and aftertreatment diesel particulate filter (DPF), in conjunction with the information provided within the procedures listed in Table 29 below. See the corresponding Service Manual.
- The reuse guidelines contained in this document are the **only** Cummins Inc. approved inspection methods and pass/fail criteria with regards to inspecting an aftertreatment DPF for reuse. Many aftertreatment DPF cleaning machine manufacturers provide pass/fail criteria for aftertreatment DPFs. Pass/fail criteria based on flow-through resistance characteristics do **not** align with Cummins® aftertreatment DPF reuse guidelines found in this bulletin and are **not** to be used to determine if a Cummins® aftertreatment DPF is acceptable for reuse.

Note : The Euro VI product operates an exchange program for the DPF in the event where the maintenance interval is reached. The filter will undergo a core acceptance

inspection following the guidelines set out in the ReCon® handbook. See link provided.

Note : Please refer to the “Filter Modules – Particulate – Other – 2010 and Future MR/HD” section in the ReCon® Core Acceptance Handbook (<https://quickserve.cummins.com/qs3/portal/service/manual/en/4360362/>)

II. Reuse Guidelines Summary

Note : All DPF reuse guidelines apply to both DPF types unless otherwise stated. See Figure 1 and Table 1 below for DPF type identification.

Aftertreatment Diesel Oxidation Catalyst (DOC) and Diesel Particulate Filter (DPF) Reuse Guidelines

Component	Corresponding Table and Figure	Condition	Reuse Statement	Additional Remarks
DOC Outlet	2	New with writing on face	Unconditional Reuse	See Table 2 below.
DOC Outlet	6	Light amounts of soot on face	Unconditional Reuse	See Table 6 below.
DOC Outlet	7	Dimpling or small chipped sections.	Conditional Reuse	See Table 7 Below
DOC Inlet	8	Face plugged	Conditional Reuse	See Table 8 below.
DOC Inlet or Outlet	9	Melted or Deformed Cells	Do Not Reuse	See Table 9 below.
DPF (Type A)	3	New with writing on face	Unconditional Reuse	See Table 3 below.
DPF (Type A) ReCon®	4	New with pink face	Unconditional Reuse	See Table 4 below.
DPF (Type B)	5	New with writing on face	Unconditional Reuse	See Table 5 below.
DPF Inlet	10	After cleaning very little soot on face or within channels	Unconditional Reuse	See Table 10 below.
DPF Inlet	11	Light levels of soot on face and within channels	Unconditional Reuse	See Table 11 below.

Aftertreatment Diesel Oxidation Catalyst (DOC) and Diesel Particulate Filter (DPF) Reuse Guidelines				
Component	Corresponding Table and Figure	Condition	Reuse Statement	Additional Remarks
DPF Inlet	12	Moderate levels of soot on face and within channels	Unconditional Reuse	See Table 12 below.
DPF Inlet	13	Light levels of soot on face and within channels	Unconditional Reuse	See Table 13 below.
DPF Inlet	14	Moderate levels of soot on face and within channels	Unconditional Reuse	See Table 14 below.
DPF Inlet	15	Heavy levels of soot with face plugging	Conditional Reuse	See Table 15 below.
DPF Inlet	16	Concentrated white ash, dark wet surrounding ring, and uneven soot streaks across inlet	Do Not Reuse	See Table 16 below.
DPF Inlet	17	Dark soot rings	Conditional Reuse	See Table 17 below.
DPF Outlet	18	Dark streak on face and coolant staining	Conditional Reuse	See Table 18 below.
DPF Outlet	19	Black soot stains on filter outlet and in filter channel walls	Do Not Reuse	See Table 19 and DPF Inspection Section below.
DPF Outlet	20	Excessive black soot stains on filter outlet	Do Not Reuse	See Table 20 below.
DPF Outlet (Type B)	21	Loose segments, missing channel plugs, missing cement	Do Not Reuse	See Table 21 below.

Aftertreatment Diesel Oxidation Catalyst (DOC) and Diesel Particulate Filter (DPF) Reuse Guidelines				
Component	Corresponding Table and Figure	Condition	Reuse Statement	Additional Remarks
DPF Inlet (Type B)	22	White powdery substance present, deformed/melted appearance	Do Not Reuse	See Table 22 below.
DPF Outlet (Type A)	23	Melted or deformed cells	Do Not Reuse	See Table 23 below.
DPF Inlet and Outlet (Type B)	24	Cracking of cement lines between segments	Do Not Reuse	See Table 24 below.
DPF Inlet and Outlet (Type B)	25	Cracking of the substrate material	Do Not Reuse	See Table 25 below.
DPF Outlet (Type A)	26	Face cracked	Do Not Reuse	See Table 26 below.
DPF Inlet	27	Gouge in the face	Conditional Reuse	See Table 27 below.
DPF Inlet	28	White substance present (fibers)	Conditional Reuse	See Table 28 below.
DPF Outlet	29	Brown staining on filter outlet	Unconditional Reuse	See Table 29 below.
DPF	-	Excessive soot buildup on tailpipe and black smoke during engine operation	Do Not Reuse	See Table 19 and 20 below. See DPF Inspection Section below.
General DOC and DPF Inlet or Outlet	-	Coolant or water intrusion	Conditional Reuse	See Table 17 below.
General DOC and DPF Inlet or Outlet	-	Fuel Soaked	Do Not Reuse	See Corresponding Service Manual. Investigate cause of excessive fuel.

Aftertreatment Diesel Oxidation Catalyst (DOC) and Diesel Particulate Filter (DPF) Reuse Guidelines				
Component	Corresponding Table and Figure	Condition	Reuse Statement	Additional Remarks
General DOC and DPF Inlet or Outlet	-	Oil Soaked	Do Not Reuse	See Corresponding Service Manual. Investigate cause of excessive oil.

DOC and DPF Housing Reuse Guidelines:

- The aftertreatment DOC and aftertreatment DPF housings **must** be free of dents.
- All mounting flanges **must** be free of dents, cracks, or gouges to seal properly with gaskets.

III. Diesel Particulate Filter (DPF) Media Types

There are two types of aftertreatment DPF media. See Figure 1 and Table 1 below.

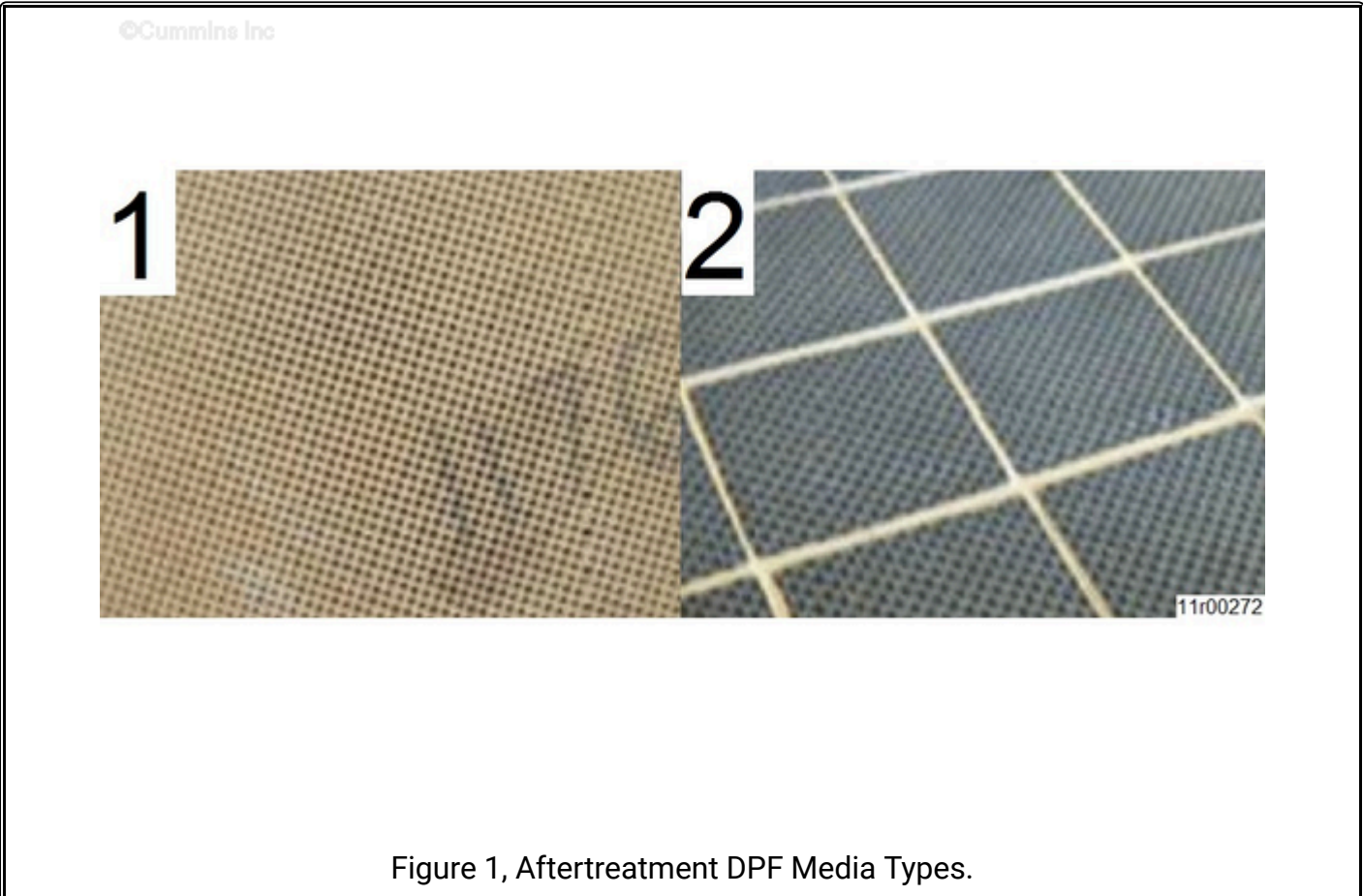


Figure 1, Aftertreatment DPF Media Types.

Figure 1:

1. Type A DPF
2. Type B DPF

Table 1, Aftertreatment Diesel Particulate Filter (DPF) Media Types	
Appearance	• (1) Type A DPF with tan media color • (2) Type B DPF with gray media color and grid face
Comment	The reuse guidelines apply to both aftertreatment DPF media types.

IV. Pictures of New Components

New Aftertreatment DOC



Table 2, New Aftertreatment Diesel Oxidation Catalyst (DOC)	
Appearance	Clean with no dents or gouges and writing on face.
Comment	Ready for use.

New Aftertreatment DPFs

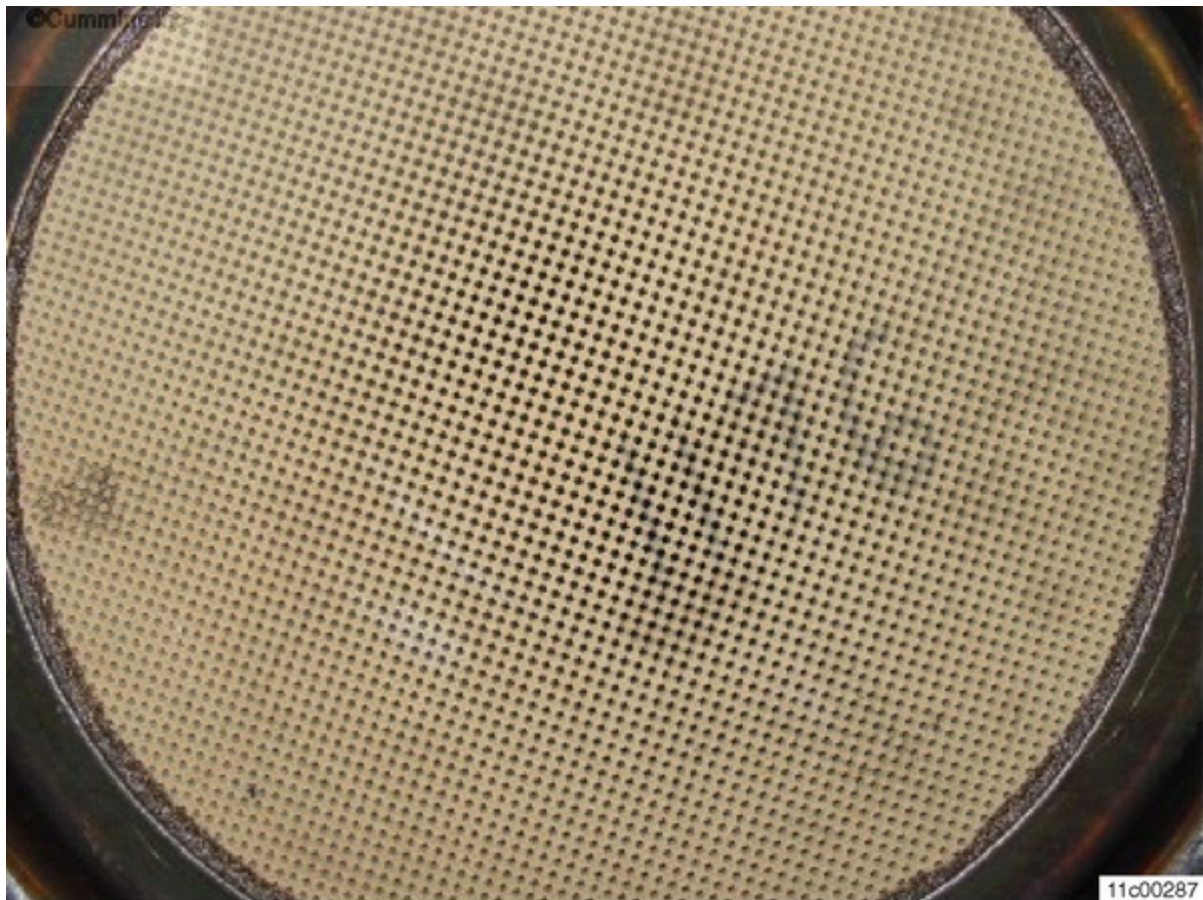


Figure 3, Aftertreatment DPF - New - (Type A).

Table 3, New Aftertreatment Diesel Particulate Filter (DPF) - (Type A)

Appearance	Clean with tan media and writing on face.
Comment	Ready for use.



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Figure 4, Aftertreatment Diesel Particulate Filter Inlet and Outlet - ReCon® - (Type A).

Table 4, New Aftertreatment Diesel Particulate Filter (DPF) - ReCon® - (Type A)

Appearance	Clean with pink media.
Action	Ready for use.

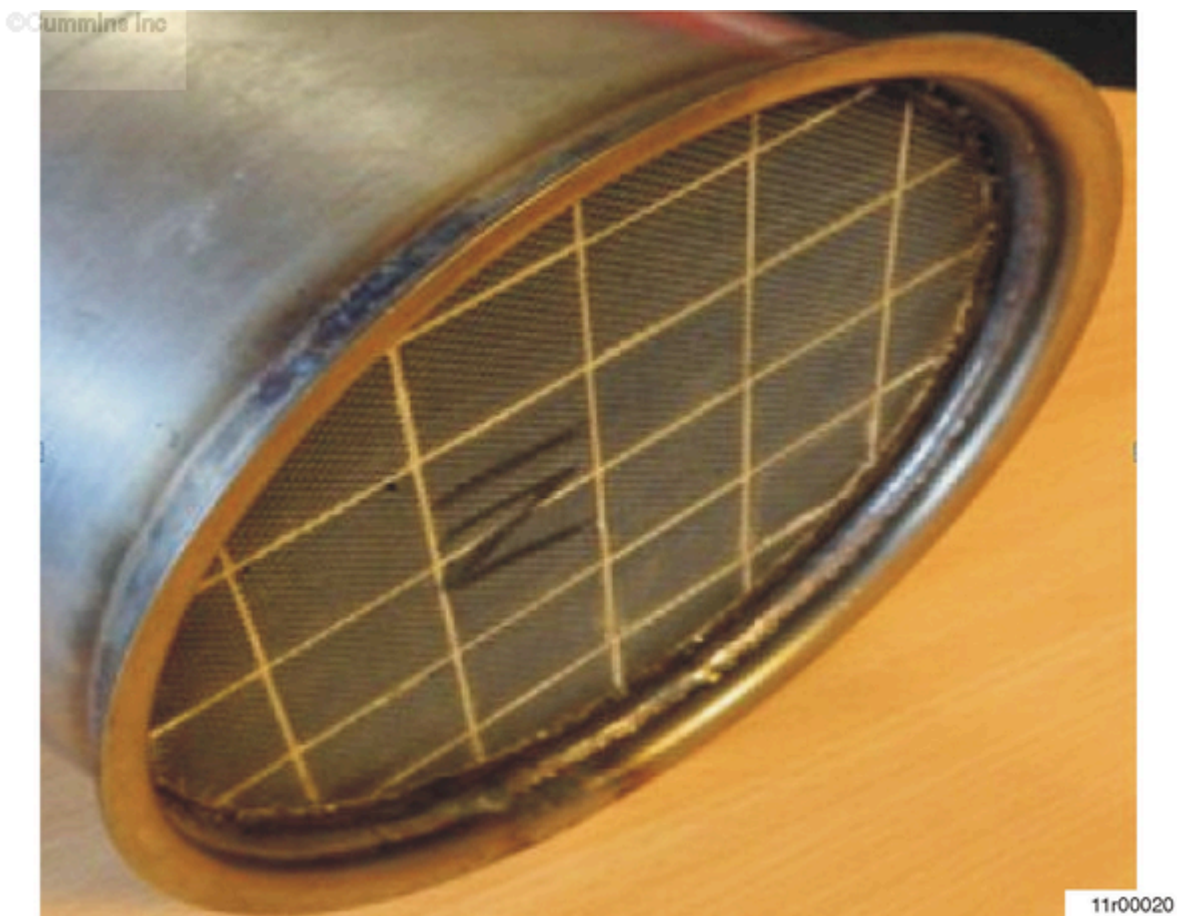


Figure 5, Aftertreatment DPF - New - (Type B).

Table 5, New Aftertreatment Diesel Particulate Filter (DPF) - (Type B)

Appearance	Clean with dark gray media, light gray jointing cement, and writing on face.
Action	Ready for use.

V. Failure Modes and Reuse Pictures

Aftertreatment Diesel Oxidation Catalyst (DOC)



Figure 6, Aftertreatment Diesel Oxidation Catalyst (DOC) Outlet - Used.

Table 6, Aftertreatment Diesel Oxidation Catalyst Outlet – Used

Appearance	Light amounts of soot on outlet face.
Cause	Normal engine operation.
Action	None.



Figure 7, Aftertreatment Diesel Oxidation Catalyst (DOC) Outlet – Dimpling/Chipped Areas.

Table 7, Aftertreatment Diesel Oxidation Catalyst Outlet – Dimpling/Chipped Areas

Appearance	Small dimple/chipped sections location along DOC outlet face.
Cause	Part handling.
Action	Confirm section is not melted. Refer to table 9 for reference.

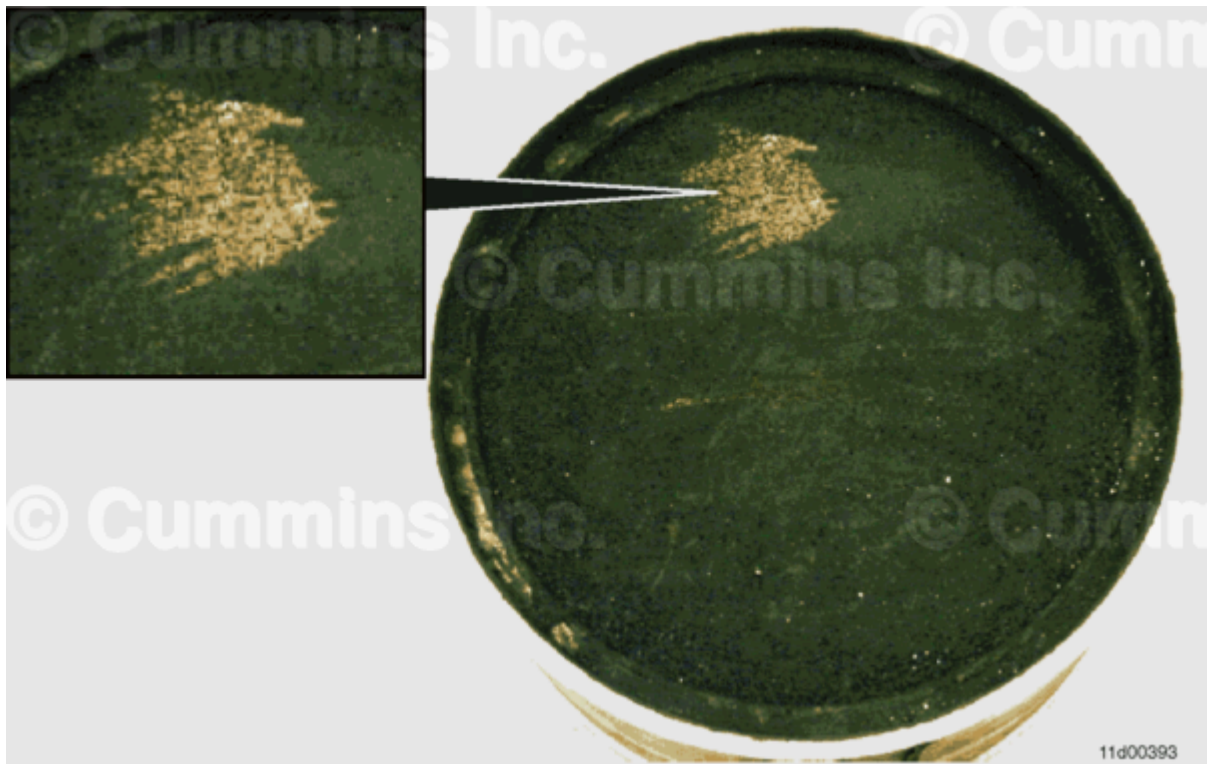


Figure 8, Aftertreatment Diesel Oxidation Catalyst (DOC) Inlet - Face Plugged.

Table 8, Aftertreatment Diesel Oxidation Catalyst (DOC) Inlet - Face Plugged

Appearance	High amounts of soot collecting on aftertreatment DOC inlet face, plugging DOC face.
Cause	A malfunction resulted in coolant entering exhaust system, light duty cycles, low exhaust temperatures, or combustion malfunction creating excessive engine smoke.
Action	Clean aftertreatment DOC. See corresponding Service Manual. Reference Procedure 011-049 in Section 11.

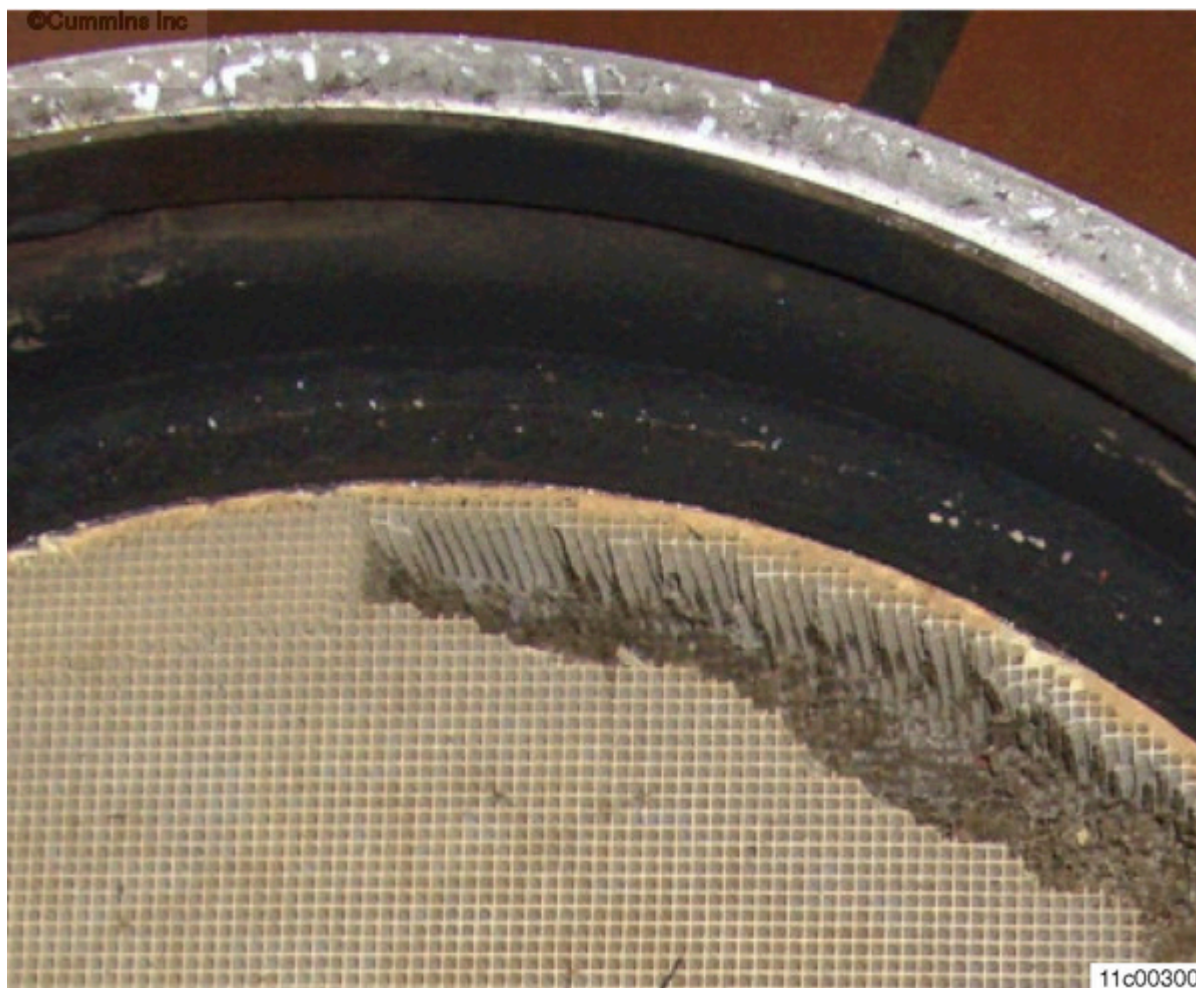


Figure 9, Aftertreatment Diesel Oxidation Catalyst (DOC) - Melted.

Table 9, Aftertreatment Diesel Oxidation Catalyst (DOC) - Melted

Appearance	Melted or deformed cells.
Cause	Excessive fuel or oil in exhaust.
Action	• Check for fault codes related to aftertreatment system. • Investigate engine service history. • Inspect turbocharger turbine outlet and aftertreatment inlet for oil or fuel. • Only replace aftertreatment DOC after cause of excessive black smoke, fuel, or oil in exhaust has been corrected.

Aftertreatment Diesel Particulate Filter (DPF)

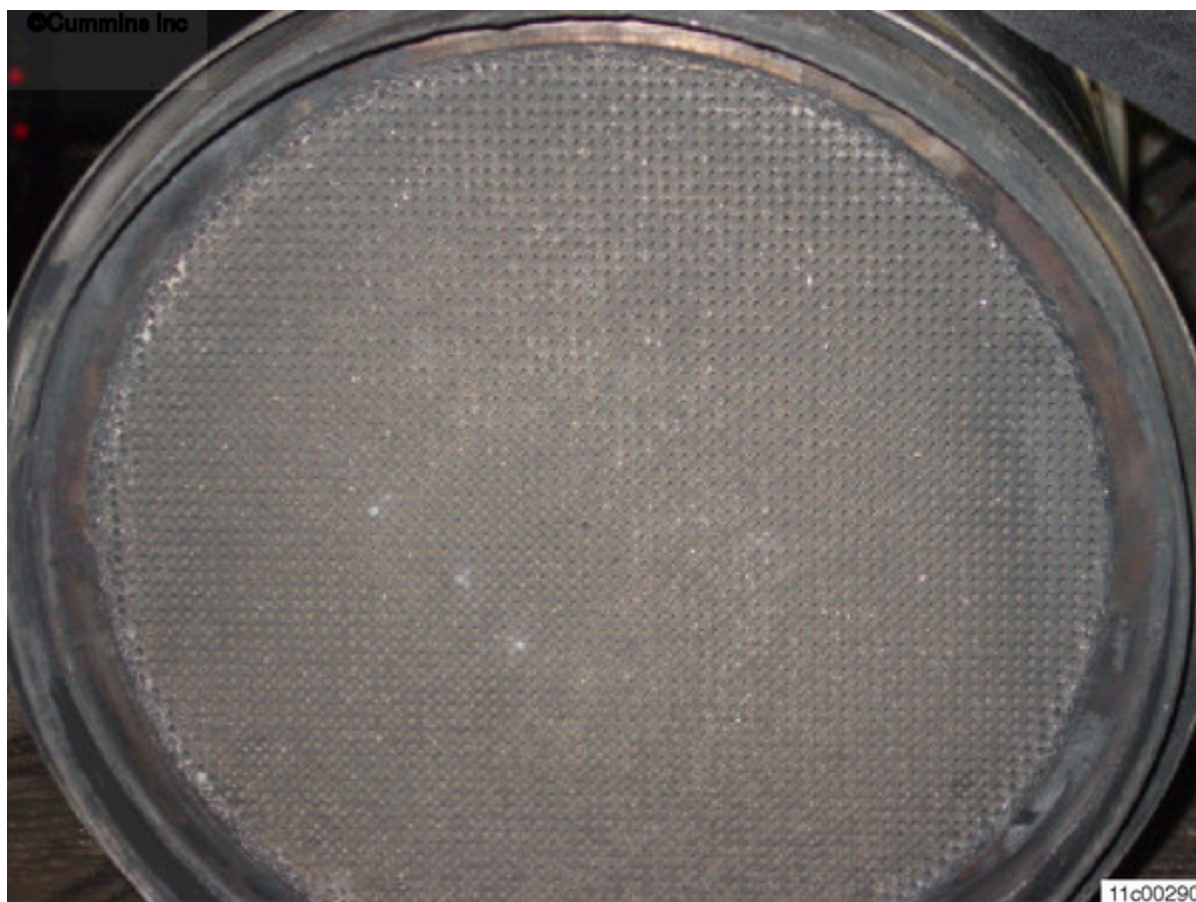


Figure 10, Aftertreatment Diesel Particulate Filter (DPF) - Inlet - After Cleaning.

Table 10, Aftertreatment Diesel Particulate Filter (DPF) Inlet - After Cleaning

Appearance	Very little soot (black) or ash (gray) on face or within channels.
Cause	Normal engine operation followed by a cleaning by aftertreatment DPF cleaning machine.
Action	Ready for use.



Figure 11, Aftertreatment Diesel Particulate Filter (DPF) Inlet - Light Soot Load.

Table 11, Aftertreatment Diesel Particulate Filter (DPF) Inlet - Light Soot Load (Type A)

Appearance	Light levels of soot (black) or ash (gray) on face and within channels.
Cause	Normal engine operation.
Action	None, can continue to be used.



Figure 12, Aftertreatment Diesel Particulate Filter (DPF) Inlet - Moderate Soot Load.

Table 12, Aftertreatment Diesel Particulate Filter (DPF) Inlet - Moderate Soot Load (Type A)

Appearance	Moderate levels of soot (black) or ash (gray) on face and within channels.
Cause	Normal engine operation.
Action	None, can continue to be used.

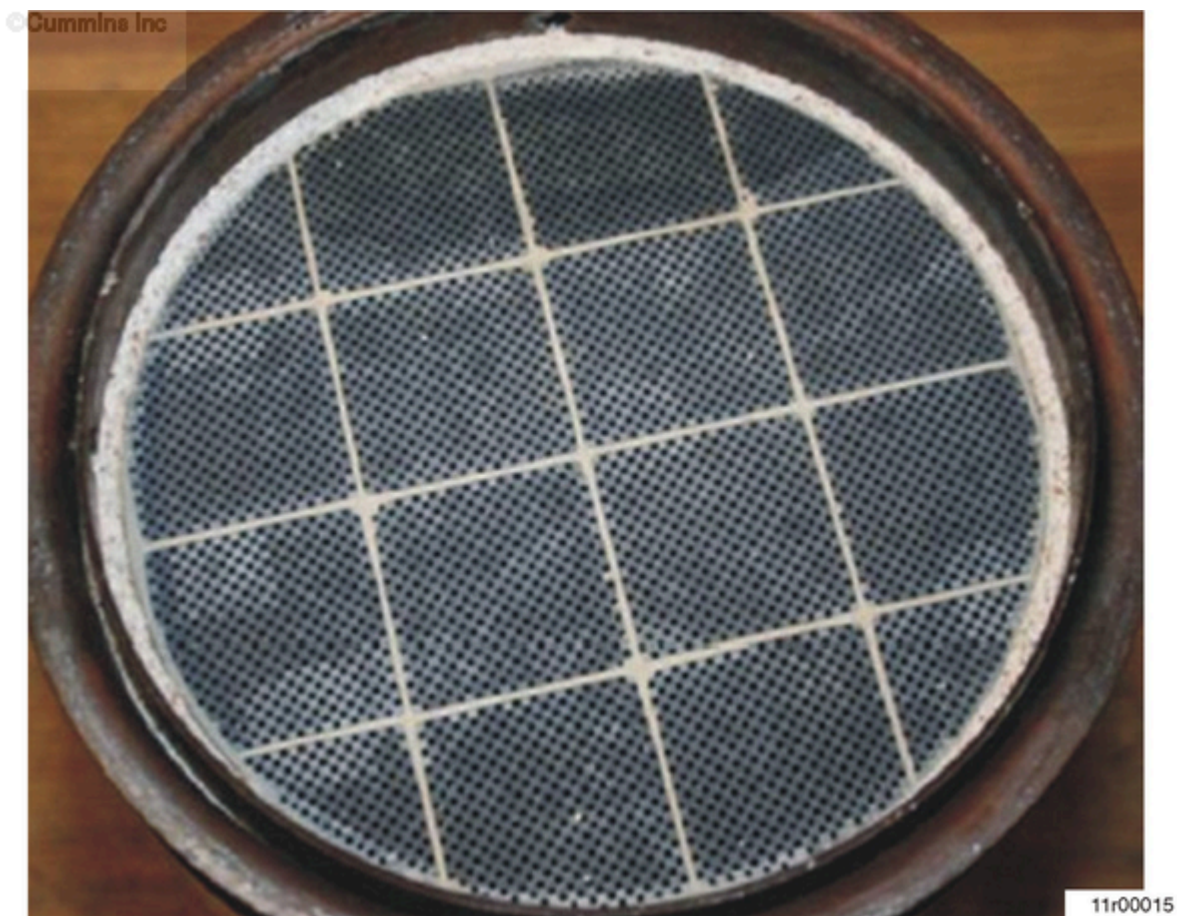


Figure 13, Aftertreatment Diesel Particulate Filter (DPF) Inlet - Light Soot Load.

Table 13, Aftertreatment Diesel Particulate Filter (DPF) Inlet - Light Soot Load (Type B)

Appearance	Light levels of soot (black) or ash (gray) on face and within channels.
Cause	Normal engine operation.
Action	None, can continue to be used.

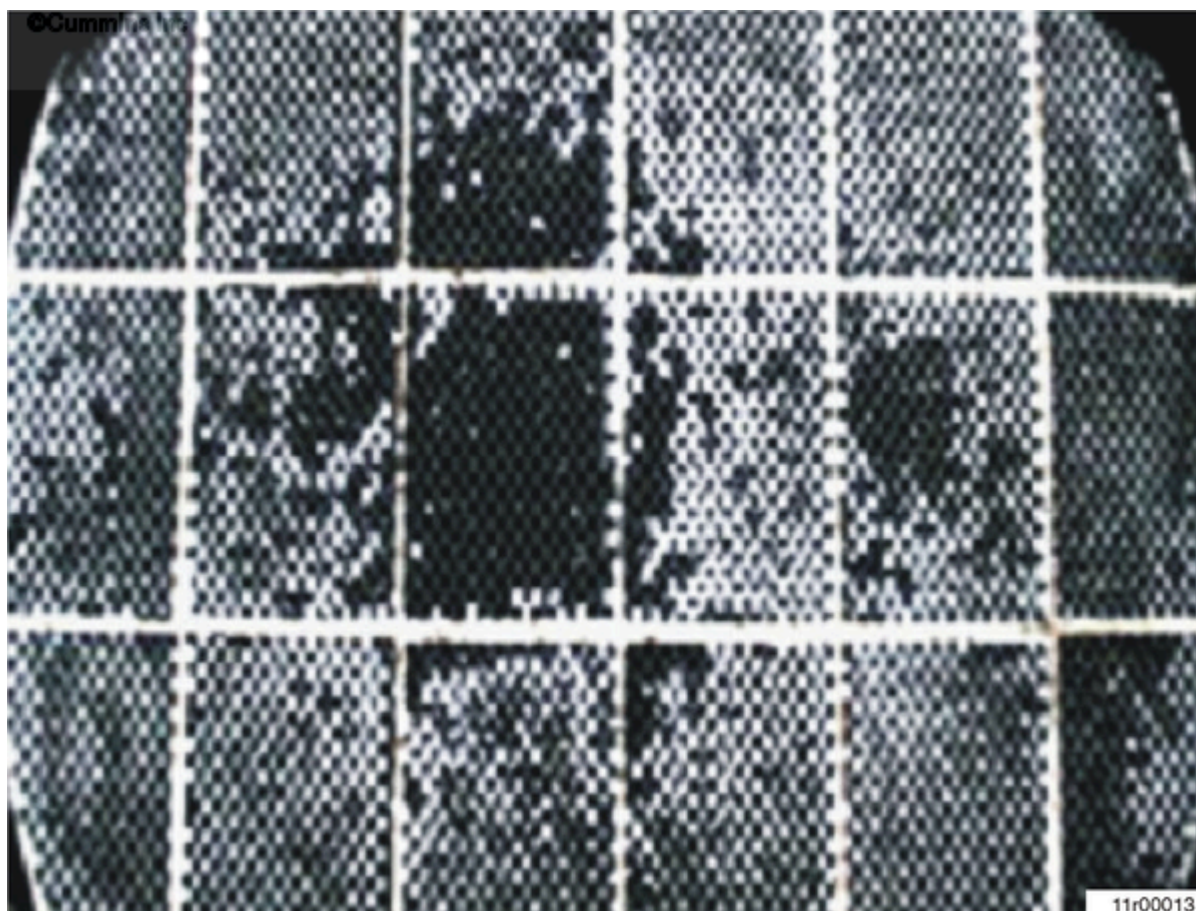


Figure 14, Aftertreatment Diesel Particulate Filter (DPF) Inlet - Moderate Soot Load.

Table 14, Aftertreatment Diesel Particulate Filter (DPF) Inlet - Moderate Soot Load (Type B)

Appearance	Moderate levels of soot (black) or ash (gray, white) on face, but not blocking channel openings.
Cause	Normal engine operation.
Action	None, can continue to be used.

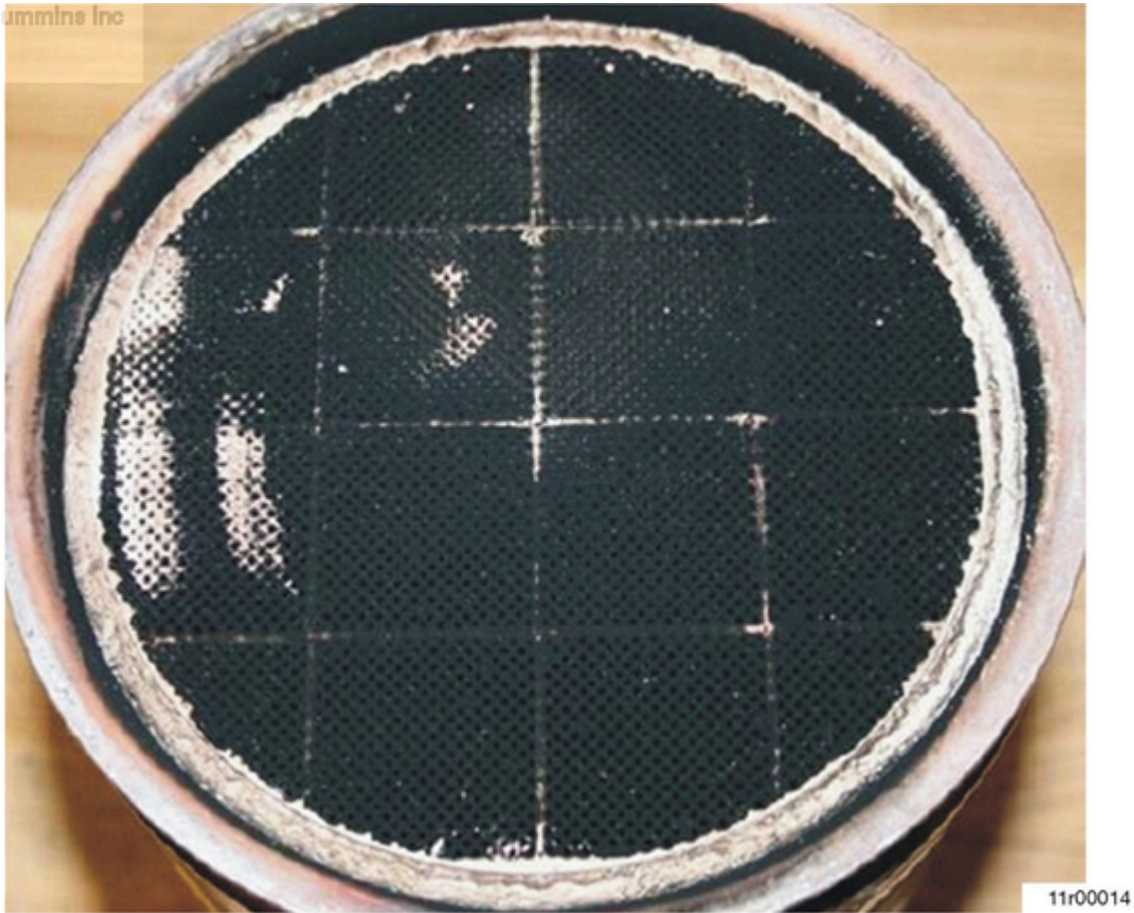


Figure 15, Aftertreatment Diesel Particulate Filter (DPF) - Inlet - Heavy Soot Load.

Table 15, Aftertreatment Diesel Particulate Filter (DPF) Inlet - Heavy Soot Load

Appearance	Heavy levels of soot load (black) with face plugging (channel openings narrowed or blocked).
Cause	Normal use.
Action	Check for active fault codes, Follow troubleshooting steps for soot related fault codes. Many engine models do not recommend DPF air cleaning. Return the aftertreatment DPF section to a Cummins® Authorized Repair Location for exchange, replacement, or cleaning according to the corresponding Service Manual and/or Fault Code Troubleshooting Manual.

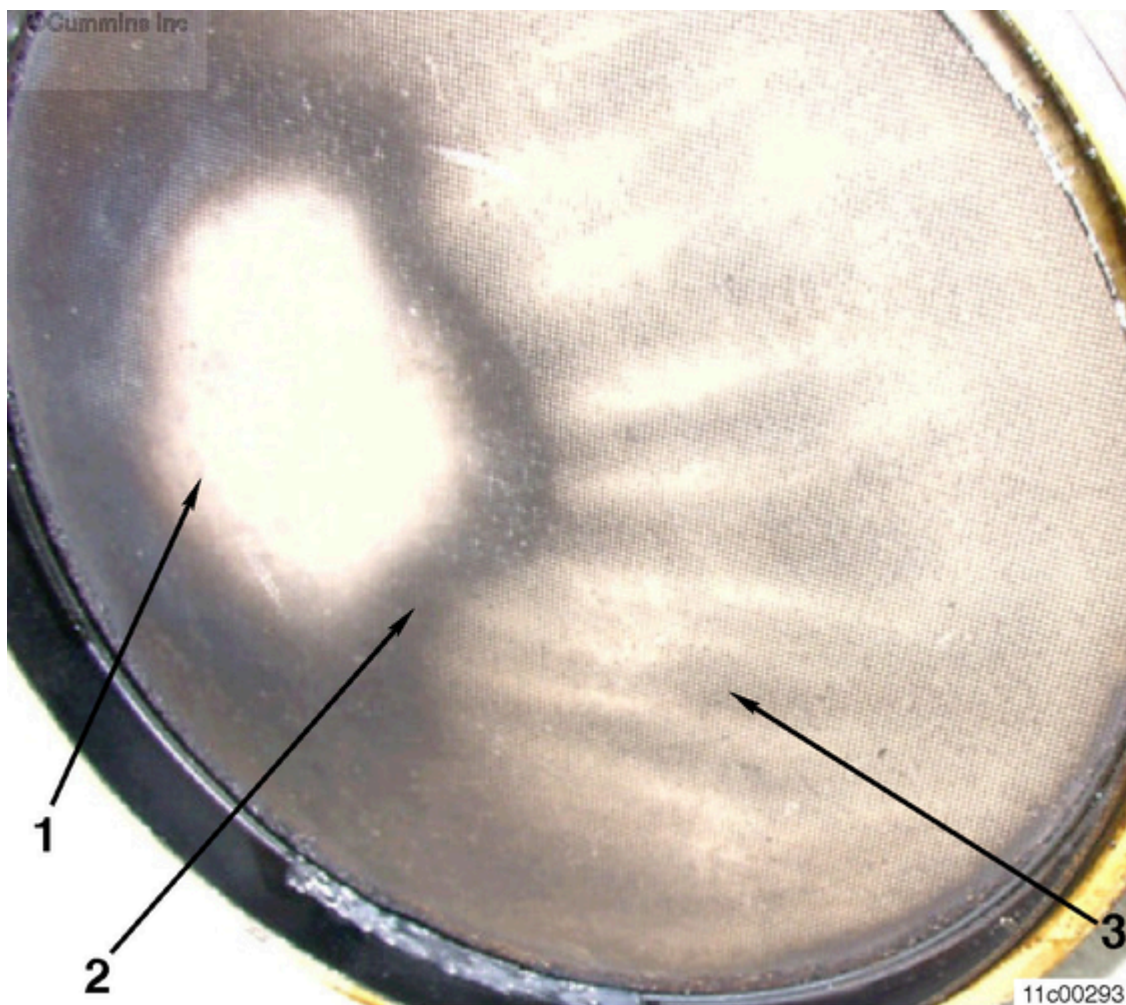


Figure 16, Aftertreatment Diesel Particulate Filter (DPF) - Inlet - Ash.

Figure 16:

1. Concentrated white ash
2. Dark wet surrounding ring
3. Uneven soot streaks

Table 16, Aftertreatment Diesel Particulate Filter Inlet – Ash

Appearance	Concentrated white ash (Figure 16, Item 1), dark wet surrounding ring (Figure 16, Item 2), and uneven soot streaks (Figure 16, Item 3) across inlet.
Cause	Excessive lubricating oil concentration in exhaust gases.

Table 16, Aftertreatment Diesel Particulate Filter Inlet – Ash

Action

- Investigate cause of excessive lubricating oil in exhaust gases. Inspect turbocharger turbine outlet and aftertreatment inlet for lubricating oil.
- Investigate vehicle for engine component damage that has introduced excessive lubricating oil into exhaust. The lubricating oil contamination source **must** be corrected before replacing aftertreatment DPF.
- Aftertreatment **DPF** must be replaced.

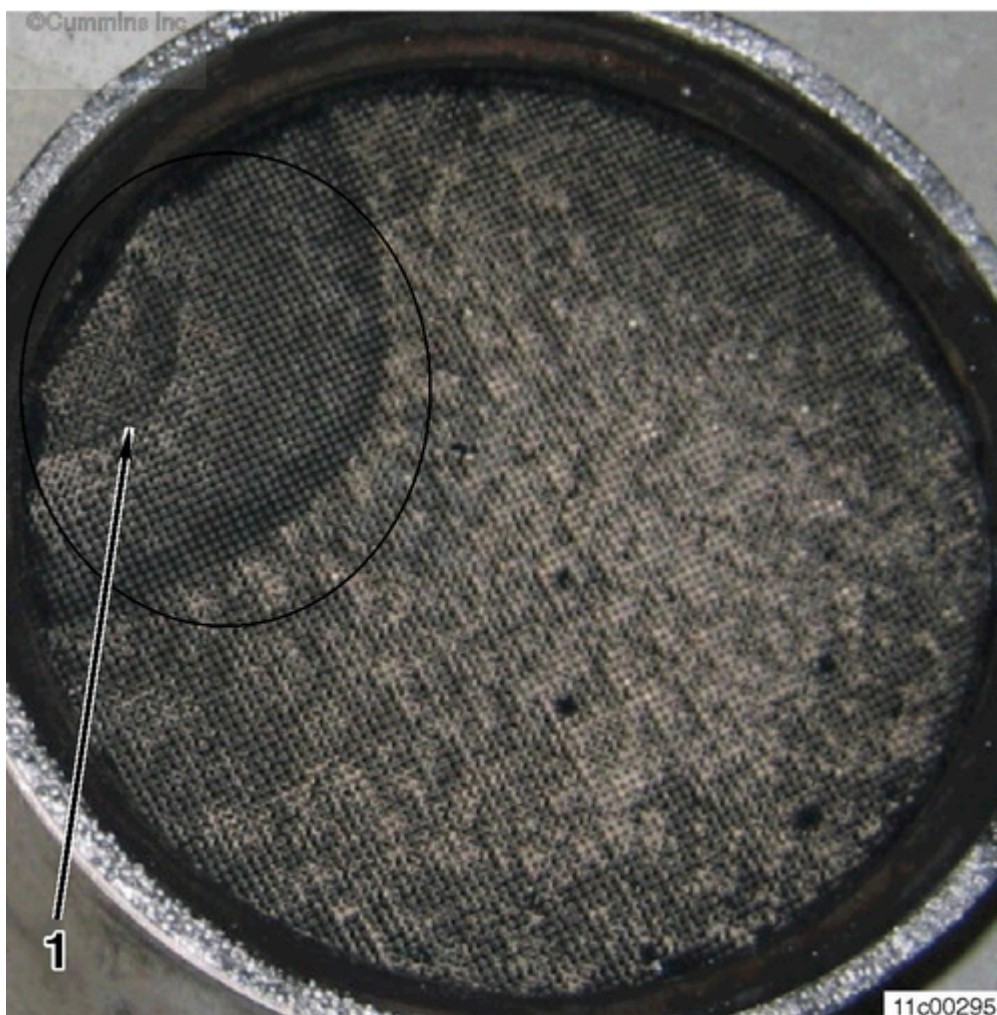


Figure 17, Aftertreatment Diesel Particulate Filter (DPF) Inlet - Dark Soot Rings.

Figure 17:

1. Dart Soot Rings

Table 17, Aftertreatment Diesel Particulate Filter (DPF) Inlet - Dark Soot Rings	
Appearance	Dark soot rings (Figure 17, Item 1).
Cause	Excessive water concentration in exhaust gases.
Action	• Check for active fault codes. • Investigate coolant consumption. • Investigate vehicle history for component damage that introduces excessive coolant into exhaust. The aftertreatment DPF can be recovered. ◦ See the corresponding Service Manual. Reference Procedure 014-013 in Section 14. ◦ See the corresponding Service Manual. Reference Procedure 014-016 in Section 14.



Figure 18, Aftertreatment Diesel Particulate (DPF) Outlet - Coolant Staining.

Table 18, Aftertreatment Diesel Particulate Filter Outlet - Coolant Staining	
Appearance	Dark streaks on face and coolant staining.
Cause	Excessive coolant concentration in exhaust gases.

Table 18, Aftertreatment Diesel Particulate Filter Outlet - Coolant Staining

Action

- Check for active fault codes.
- Investigate coolant consumption.
- Investigate vehicle history for component damage that introduces excessive coolant into exhaust
- Aftertreatment DPF can be recovered.
 - See the corresponding Service Manual. Reference Procedure 014-013 in Section 14.
 - See the corresponding Service Manual. Reference Procedure 014-016 in Section 14.



Figure 19, Aftertreatment Diesel Particulate Filter (DPF) Outlet - Soot Stain.

Figure 19:

1. Soot staining in channel walls.
2. Soot staining on outlet face.

Table 19, Aftertreatment Diesel Particulate Filter (DPF) Outlet - Soot Stains	
Appearance	• Black soot stains on filter outlet and in filter channel walls (Figure 19, Item 1). This indicates the aftertreatment DPF has internal cracking or melting. • Black soot stains on filter outlet (Figure 19, Item 2). This does not indicate a malfunction has occurred. • For more information on identification, see DPF Inspection section below.
Cause	Internal cracking or melting of filter material.
Action	• Check for fault codes related to aftertreatment system. Investigate engine service history. • Determine if fuel or lubricating oil has been introduced into exhaust as the result of damage to another engine component. • Inspect turbocharger turbine outlet and aftertreatment inlet for lubricating oil or fuel. See aftertreatment DPF Inspection Section below for additional information. • Only replace aftertreatment DPF after cause of excessive black smoke, fuel, or lubricating oil in exhaust has been corrected.

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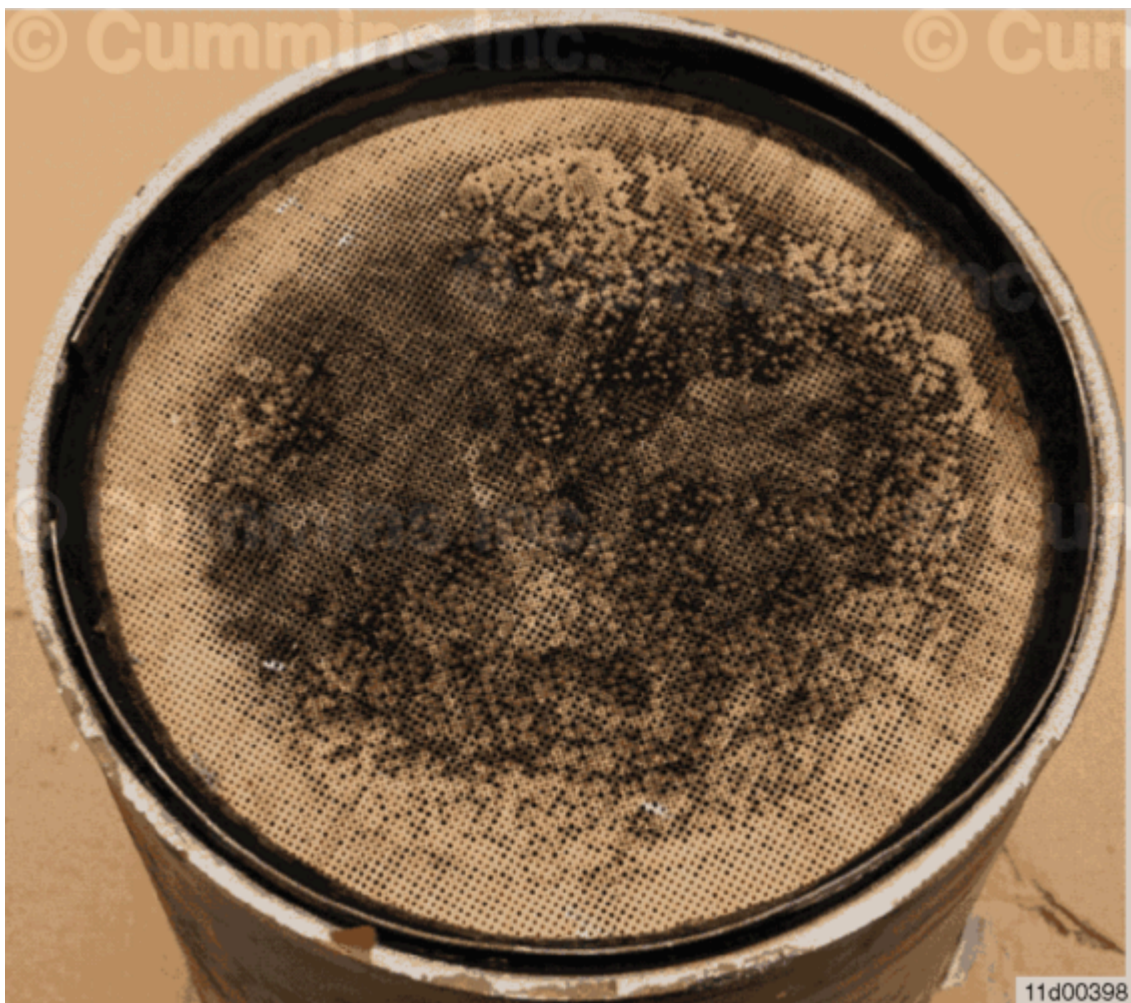


Figure 20, Aftertreatment Diesel Particulate Filter (DPF) Outlet - Excessive Soot Stains.

Table 20, Aftertreatment Diesel Particulate Filter (DPF) Outlet - Excessive Soot Stains

Appearance	Excessive black soot stains on filter outlet.
Cause	Internal cracking or melting of filter material.
Action	• Check for fault codes related to aftertreatment system. • Investigate engine service history. • Inspect turbocharger turbine outlet and aftertreatment inlet for lubricating oil or fuel. • Only replace aftertreatment DPF after cause of excessive black smoke, fuel, or oil in exhaust has been corrected.

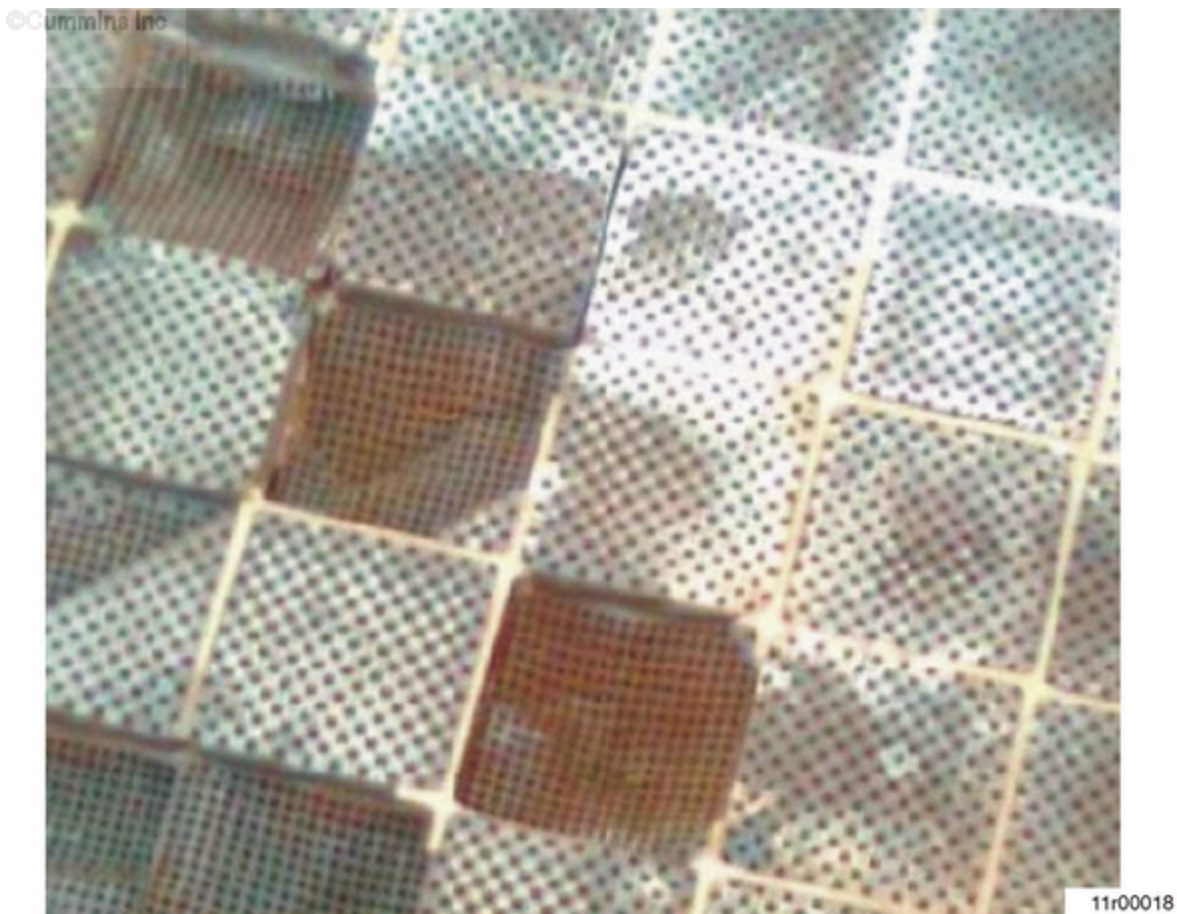
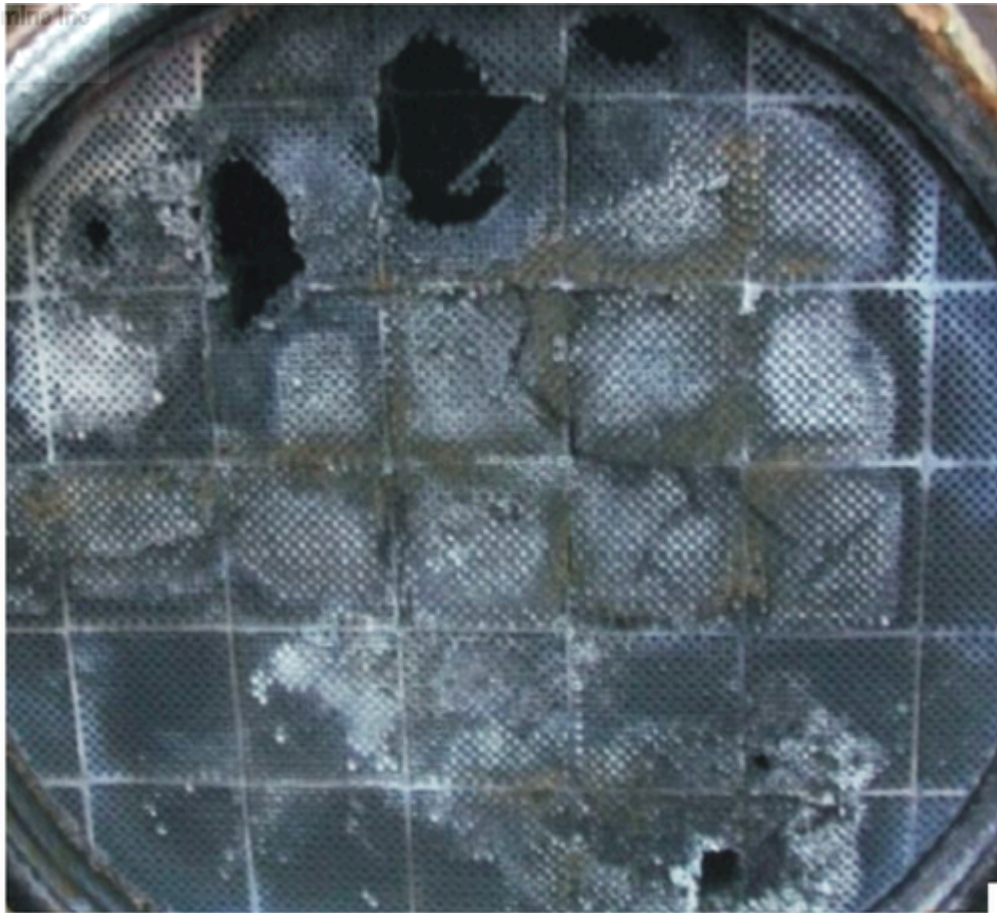


Figure 21, Aftertreatment Diesel Particulate Filter (DPF) Outlet – Thermal Damage - (Type B).

Table 21, Aftertreatment Diesel Particulate Filter (DPF) Outlet - Thermal Damage - (Type B)

Appearance	Loose segments, missing channel plugs, and missing cement.
Cause	Thermal damage.
Action	Replace aftertreatment DPF.



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Figure 22 , Aftertreatment Diesel Particulate Filter (DPF) Inlet - Melted - (Type B).

Table 22, Aftertreatment Diesel Particulate Filter (DPF) Inlet - Melted - (Type B)

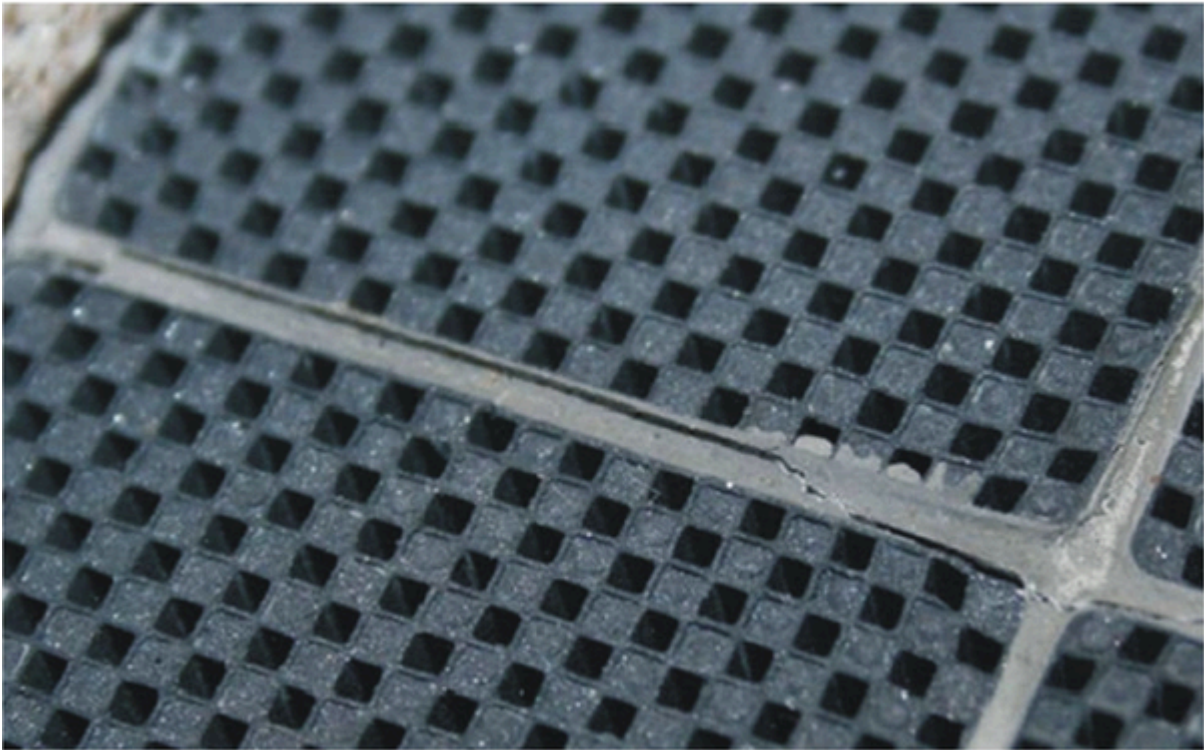
Appearance	White powdery substance present, deformed/melted appearance.
Cause	Thermal damage caused by excessive amounts of oil or fuel entering exhaust stream and saturating DPF.
Action	• Check for fault codes related to aftertreatment system. • Investigate engine service history. • Determine if fuel or oil has been introduced into exhaust from damage of another engine component. • Inspect turbocharger turbine outlet and aftertreatment inlet for oil or fuel. • Only replace aftertreatment DPF after cause of excessive fuel or oil in exhaust has been corrected.



Figure 23, Aftertreatment Diesel Particulate Filter (DPF) Outlet – Melted - (Type A).

Table 23, Aftertreatment Diesel Particulate Filter (DPF) Outlet - Melted - (Type A)

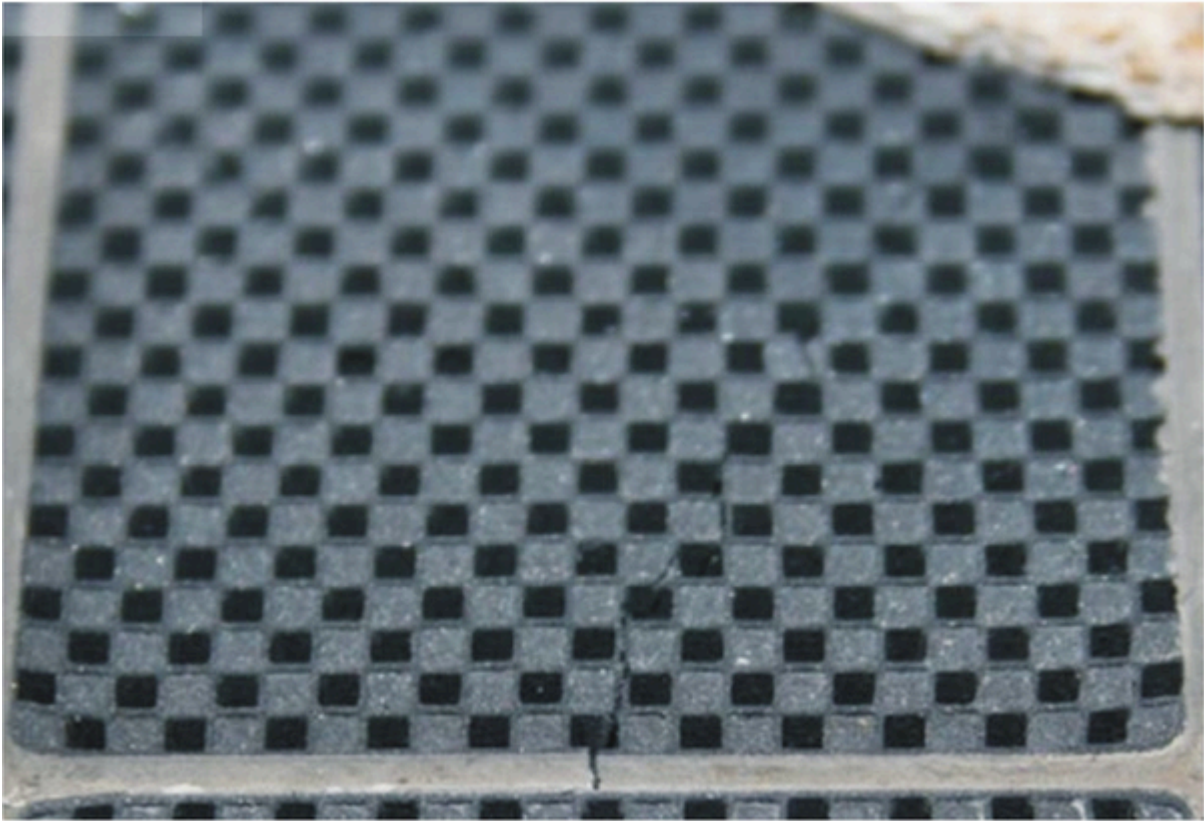
Appearance	Melted or deformed cells.
Cause	Excessive fuel or oil in exhaust.
Action	• Check for fault codes related to aftertreatment system. • Investigate engine service history. • Determine if fuel or oil has been introduced into exhaust from damage of another engine component. • Inspect turbocharger turbine outlet and aftertreatment inlet for oil or fuel. • Only replace aftertreatment DPF after cause of excessive fuel or oil in exhaust has been corrected.



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Figure 24 , Aftertreatment Diesel Particulate Filter (DPF) Inlet and Outlet – Cracking - (Type B).

Table 24, Aftertreatment Diesel Particulate Filter (DPF) Inlet and Outlet - Cracking - (Type B)	
Appearance	Cracking of cement lines between segments.
Cause	Mishandling, impact damage.
Action	Replace aftertreatment DPF.



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Figure 25, Aftertreatment Diesel Particulate Filter (DPF) Inlet and Outlet – Cracking - (Type B).

Table 25, Aftertreatment Diesel Particulate Filter (DPF) Inlet and Outlet– Cracking - (Type B)	
Appearance	Cracking of filter material.
Cause	Mishandling, impact damage.
Action	Replace aftertreatment DPF.



Figure 26, Aftertreatment Diesel Particulate Filter (DPF) Outlet - Face Cracked - (Type A).

Table 26, Aftertreatment Diesel Particulate Filter (DPF) Outlet - Face Cracked - (Type A)	
Appearance	Crack across filter face.
Cause	High soot load in aftertreatment DPF reacted rapidly with engine exhaust, creating a high temperature.
Action	Replace aftertreatment DPF.



Figure 27, Aftertreatment Diesel Particulate Filter (DPF) Inlet – Gouged.

Table 27, Aftertreatment Diesel Particulate Filter (DPF) Inlet - Gouged

Appearance	Gouge in filter face.
Cause	Face gouge created during manufacturing of filter, or an object scraped across filter face.
Action	• Check if gouge(s) in substrate material is more than 3 mm [1/8 in] deep. • If gouge(s) is deeper than 3 mm [1/8 in], more than 20 cell plugs are missing or damaged from gouge(s), replace aftertreatment DPF.



Figure 28, Aftertreatment Diesel Particulate Filter (DPF) Inlet - Fibers from Insulation.

Table 28, Aftertreatment Diesel Particulate Filter (DPF) Inlet - Fibers from Insulation

Appearance	White substance present.
Cause	Fibers from upstream component.
Action	Aftertreatment DPF needs cleaning through recommended process, if applicable.

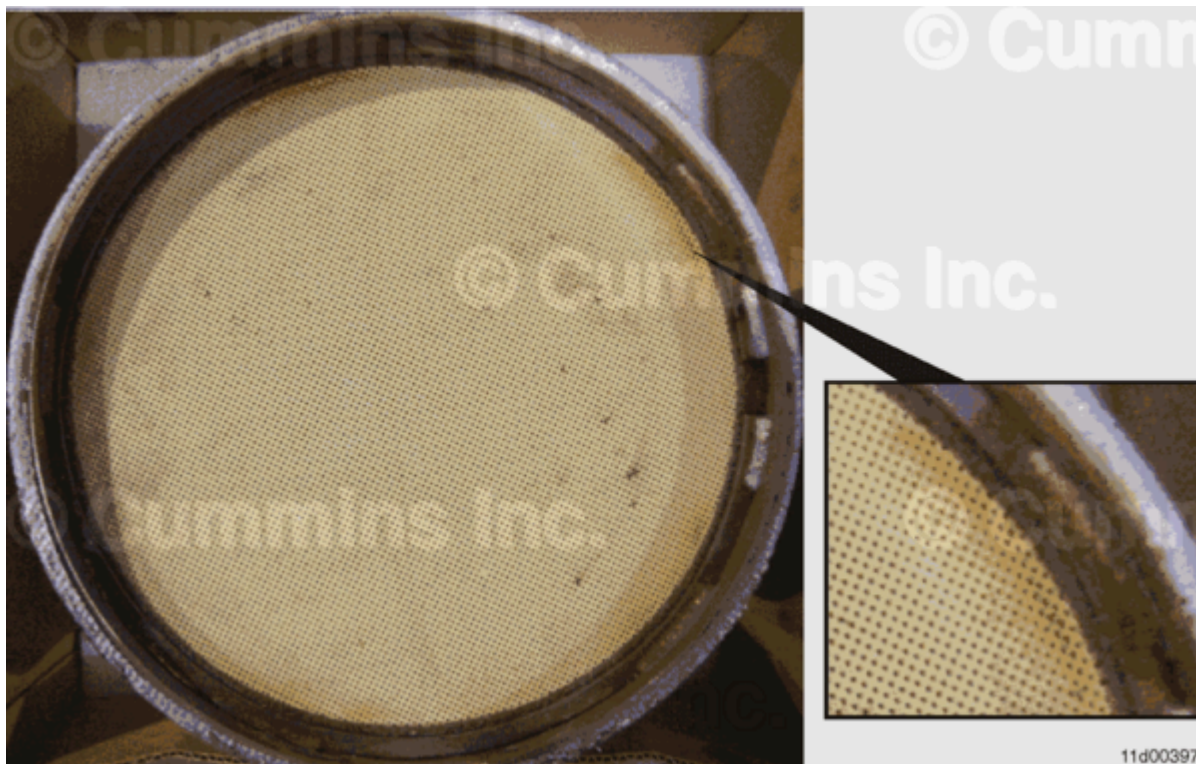


Figure 29, Aftertreatment Diesel Particulate Filter (DPF) Outlet - Burn Marks from Welding Process.

Table 29, Aftertreatment Diesel Particulate Filter (DPF) Outlet - Burn Marks from Welding Process

Appearance	Brown staining on filter outlet.
Cause	Welding process during assembly of filter.
Action	Inspect filter for black soot stains in outlet cell walls. If only brown burn marks are visible, filter can be reused.

VI. Aftertreatment Diesel Particulate Filter (DPF) Inspection

An internally cracked or melted aftertreatment DPF can show damage in three ways:

1. Aftertreatment DPF outlet channels have soot stains
 - Stains typically will **not** show on face of DPF, but can be seen in non-plugged channel walls of DPF outlet. See Figure 30 below for example.
 - Other staining or marks on face of DPF are not indications of a cracked or melted DPF. See Figure 30 below for example.
2. Exhaust system outlet has excessive soot buildup
 - Normal operation of vehicle over its life will cause small amount of soot staining to appear in outlet of exhaust, but excessive soot is an indication of a damaged DPF.
 - If excessive soot is found in outlet of exhaust and DPF is replaced, exhaust outlet must be cleaned.
3. Black smoke during engine operation

- Complaint to be verified by performing Snap Acceleration Test - Aftertreatment Connected.

Repair Direction:

- If DPF outlet channels have soot stains (1); replace DPF.
- If exhaust system outlet has excessive soot buildup (2) and there is black smoke during engine operation (3); replace DPF. The source of excessive soot that caused the cracking or melting **must** be investigated.

Additional Information:

There are two reasons aftertreatment DPF outlet channels would **not** have soot staining, but unit would have excessive soot buildup on exhaust outlet and black smoke during engine operation:

1. If an active regeneration was performed shortly before the DPF was removed and inspected, this could have removed the soot from the channels of the DPF outlet.
2. If the cracking or melting occurs close to the DPF inlet face, the soot will **not** build-up in the outlet channel wall and will **not** be visible. See Figure 30 below.

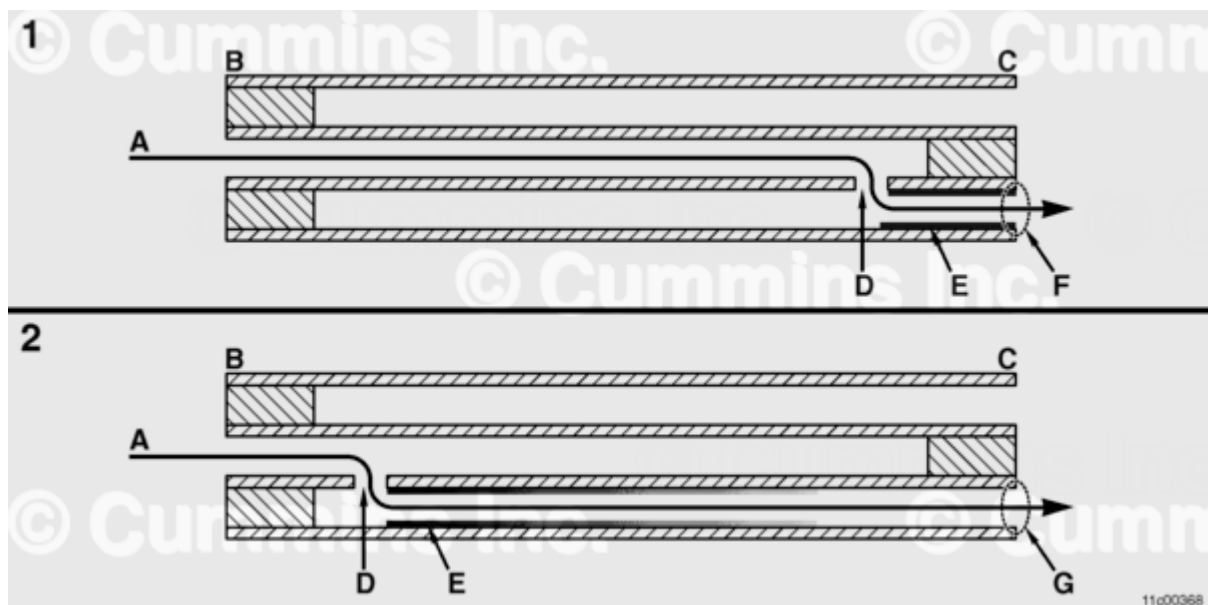


Figure 30, Mechanism of Darkened Cell in Relation to Position of Breach.

Figure 30:

1. Breach exists near outlet face
 2. Breach exists near inlet face
- A. Exhaust gas containing soot
 B. Inlet face
 C. Outlet face
 D. Breach
 E. Soot leakage
 F. Darkened
 G. **Not** darkened

VII. Associated Procedures

Table 30, Associated Procedures

Manual Type	Procedure Title	Procedure	Section
Service Manual	Aftertreatment Diesel Particulate Filter	011-041	11
	Aftertreatment Diesel Oxidation Catalyst	011-049	11
	Aftertreatment Diesel Particulate Filter Restriction Test	011-085	11
	Aftertreatment Testing	014-013	14
	Aftertreatment Diesel Particulate Filter (DPF) Regeneration Test	014-016	14
	Snap Acceleration Test	014-017	14

Document History

Date	Details
xxxx-xx-xx	Module Created
2011-9-21	Changed reference format
2011-11-28	Aftertreatment Diesel Oxidation Catalyst and Aftertreatment Diesel Particulate Filter Reuse Guidelines, has several notes that are incorrectly placed. This makes the document confusing, this edit is to reposition several of the notes to improve the document flow.
2012-2-6	Ticket Number: 36356 Reason: Hyperlink Error Notes: none
2012-2-8	Ticket Number: 36461 Reason: Spelling Error Notes: none
2012-2-8	QSOL Quick Fix Reason: Spelling Error Notes: none
2012-4-3	Update for new HD Industrial and Power Generation product.
2012-4-24	update
2012-5-31	QSOL Quick Fix Reason: Outfile Notes: none

Date	Details
2012-7-6	Add statement allowing the use of the aftertreatment diesel particulate filter restriction test on ISX15 and QSX15 to evaluate DPF for reuse when it is suspected to be ash plugged.
2013-5-3	Add DPF Restriction Test for Heavy Duty engines and associate the 2013 engines with this manual.
2013-5-13	QSOL Quick Fix Reason: Outfile Notes: none
2013-5-13	QSOL Quick Fix Reason: Outfile Notes: none
2013-8-19	Update is required due to the introduction of a new DPF substrate material.
2014-3-3	WO 58698
2014-8-26	none
2014-9-10	none
2014-9-29	Add QSX15 CM2350 and QSG12 CM2350
2014-11-11	Fix broken link
2015-12-14	Add ISV5.0 to the Associated Procedures Table. Added Recon® DPF.
2016-4-26	Fix broken link. Reformatted for readability.
2016-8-17	Reformatted document.
2017-4-13	Updated Table 14 Action step.
2017-12-7	Added Figure 6: 11r00328.
2017-12-11	Updated Figure 6.
2018-6-13	Repaired link to ReCon® handbook.
2018-6-14	Repaired link to ReCon® handbook.
2019-3-4	updated reference format for figures
2020-11-10	Updated action in Table 15.